

Fig. 4: Actual-size PCB of the home theatre backlight unit

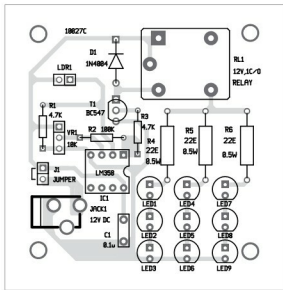


Fig. 5: Component layout of the PCB shown in Fig. 4

PARTS LIST

Semiconductors:

- IC1 - LM358 op-amp
- T1 - BC547 npn transistor
- D1 - 1N4004 rectifier diode
- D2 - BAT85 Schottky diode
- D3 - 1N4148 signal diode
- LED1-LED9 - 5mm white/blue LEDs

Resistors (all 1/4-watt, 45% carbon, unless stated otherwise):

- R1, R3 - 4.7k-ohm
- R2 - 100k-ohm
- R4-R6 - 22-ohm, 0.5W

- VR1 - 10k-ohm potmeter

Capacitors:

- C1 - 0.1μF ceramic disk
- C2 - 100μF, 16V electrolytic

Miscellaneous:

- JACK1 - 12V DC jack
- CON1 - USB-A connector
- CON2 - 2-pin connector
- RL1 - 12V, 1 C/O relay
- RL2 - 5V, 1 C/O relay
- J1 - 2-pin jumper connector
- LDR1 - Light dependent resistor (5mm)

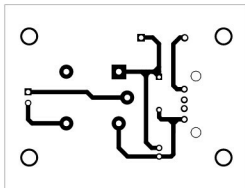


Fig. 6: Actual-size PCB of the add-on USB switch circuit

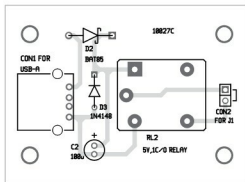


Fig. 7: Component layout of the PCB shown in Fig. 6



Fig. 8: Real-world application of the home theatre backlight unit

which, in turn, causes output of IC1 (pin 1) to go high. This activates RL1 through relay driver transistor T1 and turns on the 5mm white/blue LED strings, LED1 through LED9, connected via N/O relay contacts. Here, creation of hysteresis is achieved by adding resistor R2 in the positive feedback path.

If you want an extra AND function, that is, you want the backlight unit to work only when the television is switched on and ambient-light level is very poor, remove jumper J1 in the main circuit and attach the add-on circuit (USB switch shown in Fig. 3) at CON2. It will work as a shorting jumper when relay RL2 is energised.

Input of the add-on USB switch circuit (J1) should be connected to a vacant USB port in the television/ set-top box using a standard USB cable so that when the system is in on state, the circuit is activated by the 5V DC supply available from the USB port of the television/ set-top box.

It is very important to prevent the LDR1 to be illuminated by sources other than the natural ambient light.

Construction and testing

An actual-size, single-side PCB for the home theatre backlight unit is shown in Fig. 4 and its component layout in Fig. 5. Similarly, an actual-size, single-side PCB for the add-on USB switch circuit is shown in Fig. 6 and its component layout in Fig. 7. Enclose the PCB(s) in a small wooden box as shown in author's prototype.

LDR1, jumper J1, potmeter VR1 and other components are fitted on the cabinet. Place LDR1 in such a way that only ambient light falls on it. Connect jumper J1 externally through wire in case you are not using a USB circuit.

A picture of the real-world application example is shown in Fig. 8.

EFY notes. 1. White bright 5mm LEDs (LED1 through LED9) were used during testing.

2. The circuit can work with 3.4V/80mA, 10mm white LEDs or with 8mm blue LEDs. ●



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